Proximal Policy Optimization vs. Deep Q-Network on ALE/Venture-v5: A Comparative Study of Convergence, Stability, and Reward Exploitation under Sparse Feedback

Giovanni Vargas
20231020036

Samuel Casas
20231020038

Nahin Peñaranda
20231020032

***Abstract*—Deep Reinforcement Learning (DRL) agents face extreme challenges in environments characterised by sparse rewards and long planning horizons. This paper presents a direct empirical comparison between Proximal Policy Optimization (PPO) and Deep Q-Network (DQN) agents trained and evaluated on the ALE/Venture-v5 environment under a fixed computational budget. Building on the DQN baseline established in Challenge 1—which required a custom RAM-assisted reward shaping module to overcome gradient starvation—we implement a PPO agent equipped with Generalised Advantage Estimation (GAE), a clipped surrogate objective, entropy regularisation, and a wide convolutional actor-critic backbone (channels 64→128→128). Two PPO configurations are contrasted: a narrow baseline network (32→64→64) and the optimised wider architecture. Empirical results demonstrate that PPO achieves faster convergence, substantially higher training stability, and critically, exhibits no reward-exploitation behaviour—a systematic failure mode observed throughout DQN training. These outcomes are explained in terms of fundamental algorithmic differences between on-policy rollout learning with clipped ratio updates and off-policy temporal-difference learning with an ε-greedy replay buffer.**

***Index Terms*—Proximal Policy Optimization, Deep Q-Network, Atari 2600, Sparse Rewards, Generalised Advantage Estimation, Reward Exploitation, On-policy Learning, Arcade Learning Environment.**

I. Introduction

The design of autonomous agents capable of mastering complex sequential decision tasks from raw pixel observations has been one of the central problems in modern machine learning. Deep Q-Networks (DQN) [1] demonstrated that combining convolutional feature extraction with experience replay and fixed target networks yields human-competitive policies across a wide range of Arcade Learning Environment (ALE) [2] titles. Despite this landmark result, standard DQN architectures expose deep structural weaknesses when the environment generates rewards only after hundreds or thousands of correct actions—a condition formally identified as *sparse reward* [3].

ALE/Venture-v5, the environment assigned to Group 6 in this challenge, is a canonical member of the hard-exploration class. The player controls “Winky”, who must navigate a multi-chamber dungeon, evade or eliminate monsters, and collect treasures worth 200 points each. The game’s scoring logic creates a *reward trap*: eliminating a monster yields 0 points unless the room’s treasure has already been secured. An agent following a random policy has a statistically negligible probability of discovering a positive reward within the first million frames, driving the Q-value function into a flat, uninformative state known as *gradient starvation* [1].

In Challenge 1, a DQN agent was trained with a RAM-assisted reward shaping wrapper (`VentureRewardWrapper`) that bypassed pixel-based detection by polling emulator memory bytes `0x60`, `0x61`, `0x80`, and `0x82` to engineer exploration bonuses, treasure amplification factors, and stagnation penalties. Despite these interventions, training was characterised by slow convergence, high reward variance, and a critical failure mode: the agent repeatedly exploited the shaped reward signal by triggering the exploration bonus without completing room objectives.

Challenge 3 poses the central research question:

> *Under a fixed computational budget and on the same environment, does PPO converge faster, reach higher performance, or exhibit different failure modes compared to DQN?*

This paper answers that question through a systematic empirical study. Specifically, our contributions are:

1) A complete PPO implementation for ALE/Venture-v5 featuring GAE, clipped surrogate objective, entropy regularisation, and a theoretically motivated hyperparameter selection.
2) A controlled two-configuration ablation study isolating the effect of network width on stability and convergence in a sparse-reward environment.
3) A rigorous side-by-side comparison of DQN and PPO across four metrics: sample efficiency, asymptotic performance, training stability, and susceptibility to reward exploitation.

The remainder of this paper is organised as follows. Section II establishes the theoretical foundations for both algorithms. Section III describes the experimental pipeline. Section IV justifies the PPO hyperparameter selection. Section V presents the comparative analysis. Section VI discusses quantitative results. Section VII reports the ablation study. Section VIII concludes.

II. Theoretical Background

A. Deep Q-Networks and the Bellman Equation

DQN approximates the optimal action-value function $Q^*(s, a)$ with a neural network $Q(s, a; \theta)$ trained to minimise the temporal-difference (TD) loss:

$$L_i(\theta_i) = \mathbb{E}_{s,a,r,s'}\left[\left(y_i - Q(s,a;\theta_i)\right)^2\right], \tag{1}$$

where $y_i = r + \gamma \max_{a'} Q(s', a'; \theta^-)$ is evaluated with a periodically frozen target network θ^- [1]. Off-policy learning from a replay buffer $\mathcal{D}$ breaks temporal correlations but introduces a fundamental mismatch: the distribution of samples in $\mathcal{D}$ may differ substantially from the current policy's visitation distribution, leading to biased gradient estimates in non-stationary reward landscapes.

B. Proximal Policy Optimization

PPO [4] is an on-policy actor-critic algorithm that constrains successive policy updates to remain within a trust region, avoiding the catastrophic collapses of unconstrained gradient ascent. The clipped surrogate objective is:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\,\hat{A}_t\right)\right], \tag{2}$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio and $\hat{A}_t$ is an advantage estimate. Clipping prevents excessively large updates when $r_t(\theta)$ deviates from 1, providing first-order stability guarantees without computing second-order Hessians [4].

The combined training objective includes a value-function loss and an entropy bonus:

$$L(\theta) = -L^{\text{CLIP}} + c_1 L^{\text{VF}} - c_2 L^{\text{ENT}}, \tag{3}$$

where $L^{\text{VF}} = \mathbb{E}_t[(V_\theta(s_t) - R_t)^2]$ and $L^{\text{ENT}} = \mathbb{E}_t[H(\pi_\theta(\cdot|s_t))]$. The entropy term is especially critical in sparse-reward settings, as it actively discourages premature convergence to degenerate policies.

C. Generalised Advantage Estimation

GAE [5] interpolates between high-variance Monte-Carlo returns and high-bias one-step TD estimates using a decay parameter $\lambda \in [0, 1]$:

$$\hat{A}_t^{\text{GAE}(\gamma,\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l\, \delta_{t+l}^V, \tag{4}$$

where $\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$ is the TD residual. Setting $\lambda = 0.95$ provides a variance-bias trade-off well-suited to environments with delayed feedback [5].

D. The Sparse-Reward Dilemma in Venture

Venture's reward structure is deliberately punitive. Shooting a monster before securing the room's treasure yields exactly zero reward, penalising the most natural exploratory behaviour. For DQN, this creates a *flat gradient manifold*: $\delta_t = r_t + \gamma \max_{a'} Q(s_{t+1}, a'; \theta^-) - Q(s_t, a_t; \theta) \approx 0$ for millions of frames, effectively halting learning [1], [2]. PPO's on-policy rollouts, combined with the entropy bonus in (3), maintain stochastic exploration throughout training, generating a richer gradient signal even under reward scarcity.

III. Experimental Methodology

A. Preprocessing Pipeline

To ensure comparability, an identical preprocessing stack was applied to both DQN and PPO experiments:

1) **Grayscale conversion:** RGB channels are averaged to a single luminance channel.
2) **Spatial resizing:** Frames are scaled to 84×84 pixels using nearest-neighbour interpolation.
3) **Pixel normalisation:** Pixel values are mapped to $[0.0, 1.0]$ by dividing by 255.
4) **Frame stacking:** The last four processed observations are stacked along the channel dimension, yielding a $(4, 84, 84)$ input tensor that encodes short-term temporal dynamics.
5) **Frame skipping:** The chosen action is repeated for 4 consecutive simulator steps, reducing the inference frequency by 75% and allowing kinematics to manifest.

This pipeline follows the standardised Atari preprocessing protocol of Mnih et al. [1] and is enforced identically across all experimental conditions.

B. PPO Architecture: Two Configurations

Two convolutional actor-critic architectures were evaluated to isolate the effect of representational capacity on stability in sparse-reward environments.

Configuration 1 — Narrow Baseline Network. This network mirrors the standard DQN backbone with three convolutional layers of 32, 64, and 64 channels, followed by a 512-unit fully connected layer shared between actor and critic heads. Its representational capacity is sufficient for dense-reward games but limited in its ability to capture the complex spatial patterns required in Venture dungeons.

Configuration 2 — Optimised Wide Network (`VentureOptimizedNetwork`). Motivated by the need to detect fine-grained environmental features—room boundaries, enemy sprites, and treasure positions—this network widens the convolutional channels to 64→128→128. Two additional fully connected layers (512→512) precede separate actor and critic heads (512→256→*n_actions* and 512→256→1 respectively). All parameters are initialised using Kaiming-He normal initialisation for intermediate layers and orthogonal initialisation with small gain for the final output layers, following the best-practice recommendations of Engstrom et al. [6].

C. Training Protocol

Both PPO configurations were trained for 5,000,000 environment steps using a single GPU. A linear learning-rate schedule decays from the initial value to 10% of its initial value over the course of training, preventing oscillations near convergence.

Episode returns, per-episode dungeon counts, policy loss, value loss, and entropy are logged at every rollout boundary via TensorBoard. Best models are checkpointed whenever a new maximum episode return is achieved.

TABLE I
FINAL PPO HYPERPARAMETER CONFIGURATION

Parameter	Selected Value
Learning rate α	2.5×10^{-4}
LR schedule	Linear decay to 2.5×10^{-5}
Rollout horizon T	2,048 steps
Epochs per rollout K	4
Mini-batch size	128
Discount factor γ	0.99
GAE parameter λ	0.95
Clip epsilon ε	0.2
Entropy coefficient c_2	0.01
Value-loss coefficient c_1	0.5
Gradient norm clip	0.5
Total training steps	5,000,000
Random seed	42

The DQN baseline from Challenge 1 used 5,000,000 steps, a replay buffer of 750,000 transitions, and the RAM-assisted `VentureRewardWrapper` with shaped exploration bonuses, treasure amplification ($\times 4.5$), and stagnation penalties.

IV. HYPERPARAMETER SELECTION STRATEGY

A. Justification for Theory-Driven Selection

Venture's extreme sample complexity makes exhaustive grid search computationally infeasible. A single 5,000,000-step PPO run requires approximately 10–15 hours on commodity GPU hardware; conducting more than five independent sweeps would exceed the available budget by an order of magnitude. We therefore adopt a *theoretically grounded selection strategy* informed by large-scale empirical studies of PPO on hard Atari environments [6]–[9].

B. Final Configuration and Rationale

Table I summarises the final PPO hyperparameters.

Horizon $T = 2{,}048$. Venture's reward structure requires the agent to navigate a hallway, breach a room, and obtain a treasure before any positive signal is generated—a sequence of at minimum 300–500 actions. A horizon of $T = 512$ or $T = 1{,}024$ would frequently truncate trajectories before a reward event, producing a GAE estimate dominated by the bootstrap value rather than observed returns. Setting $T = 2{,}048$ ensures that the majority of rollouts contain at least one treasure-capture event, enabling proper credit assignment backward through the trajectory via (4) [5], [8].

Entropy coefficient $c_2 = 0.01$. In sparse-reward games, entropy collapse—the premature concentration of the policy distribution onto a small set of actions—is the on-policy analogue of DQN's gradient starvation. Maintaining $c_2 = 0.01$ provides a persistent exploration drive without overwhelming the policy gradient signal [7], [10].

Clip epsilon $\varepsilon = 0.2$. This is the canonical value from the original PPO paper [4]. A smaller value ($\varepsilon = 0.1$) was evaluated in preliminary runs and found to impede learning in early stages by preventing the policy from moving far enough to discover novel states.

GAE $\lambda = 0.95$. The value $\lambda = 0.95$ is consistently recommended for Atari tasks across multiple large-scale studies [5], [6], [9], providing a stable variance-bias trade-off for the $\gamma = 0.99$ effective horizon.

Epochs $K = 4$. Increasing K beyond 4 in sparse-reward settings risks overfitting the current rollout's advantage estimates before new environment data are collected. Cobbe et al. [11] demonstrate that excessive reuse of on-policy data introduces distribution shift analogous to the off-policy correction problem in DQN.

Network width (64→128→128). Wider convolutional filters have been shown to improve feature discrimination in visually complex Atari games by increasing the receptive field coverage at each spatial scale [12], [13]. In Venture, the ability to distinguish between open doorways, enemy sprites, and treasure objects—all rendered with similar pixel density—is directly correlated with the network's representational capacity.

V. COMPARATIVE ANALYSIS: DQN VS. PPO

A. Comparison Protocol

The comparison follows the protocol specified in the challenge guidelines [2]:

1) **Identical preprocessing:** same grayscale/resize/stack/skip pipeline.
2) **Shared evaluation metric:** mean episodic return (`ep_rew_mean`) logged at identical step intervals.
3) **Training stability:** standard deviation of episodic return over the final 100 episodes.
4) **Reward exploitation:** occurrence of systematic reward-hacking events detected through loss and reward logs.

B. Algorithmic Properties Governing Observed Behaviour

Sample efficiency. DQN's ε-greedy policy requires millions of random actions before accumulating sufficient positive transitions in the replay buffer $\mathcal{D}$. Even with the 60% exploration fraction used in Challenge 1, the agent typically required more than 3,000,000 steps before the first consistent positive gradient signal. PPO's entropy bonus maintains a stochastic policy throughout training, generating diverse trajectories from the very first rollout. Consequently, PPO reaches a meaningful learning signal—expressed as non-zero policy loss reduction—within the first 1,500,000–3,000,000 steps [4], [7].

Training stability. DQN's off-policy update rule with a fixed target network introduces instability in sparse-reward environments: when a rare positive transition finally occurs, the resulting large TD error δ_t produces a spike in the loss function, which the fixed target network cannot absorb until the next synchronisation at $\tau = 8{,}000$ steps. This manifests as the large oscillations observed in the Challenge 1 learning curves, with reward swings of several hundred points over consecutive evaluation windows. PPO's clipped objective (2) bounds the per-step policy change by design, guaranteeing that

TABLE II
QUANTITATIVE COMPARISON: DQN VS. PPO ON ALE/VENTURE-V5

Metric	DQN (Ch.1)	PPO-Narrow	PPO-Wide
Total training steps	5,000,000	5,000,000	5,000,000
First reward (steps)	>4,000,000	<4,000,000	<3,000,000
Final mean return	0–50	0–120	higher0-200
Return std. (last 100 ep)	very high	moderate	lower
Reward exploitation	Yes	No	No
RAM shaping required	Yes	No	No

each update cannot deviate catastrophically from the previous policy regardless of advantage magnitude [4], [6].

Reward exploitation. The most significant qualitative difference between the two algorithms is PPO's immunity to reward exploitation. In DQN training, the agent systematically exploited the shaped reward signal: after learning that the exploration bonus $R_{\text{room}} = +10$ could be triggered by crossing room boundaries without completing the treasure objective, the Q-function assigned high value to this oscillatory behaviour. Because DQN's replay buffer retains these transitions, the exploitative pattern was continuously reinforced across millions of steps. PPO, operating strictly on-policy with rollout data discarded after each update, does not accumulate stale exploitative transitions. Furthermore, the value critic provides a long-horizon return estimate $V(s_t)$ that directly discounts the value of boundary-crossing without subsequent collection—a structural property absent from tabular Q-value estimates under the Bellman equation (1) [4], [11].

Convergence speed. Because PPO discards each rollout after K epochs and generates fresh trajectories, the policy gradient estimated by (2) reflects the current policy's behaviour distribution throughout training. DQN's uniform replay buffer mixing over 150,000 transitions introduces a distributional lag: gradients are averaged over experiences generated by policies up to $\sim$150,000 steps in the past, slowing the propagation of recently discovered reward information [14]. In Venture, where the reward landscape changes qualitatively each time the agent masters a new room, this lag directly translates into slower adaptation.

Table II summarises the quantitative comparison between the two algorithms.

VI. QUANTITATIVE RESULTS AND DISCUSSION

A. DQN Baseline Recap (Challenge 1)

The DQN agent trained in Challenge 1 required the `VentureRewardWrapper` to produce any measurable learning signal. Without reward shaping, the agent plateaued at a mean episodic return of 0–50, confirming the gradient starvation hypothesis [1]. Even with the dense shaped reward manifold, the learning curves exhibited extremely high variance—oscillations of several hundred points per evaluation window—due to the agent discovering and subsequently exploiting the exploration bonus trigger. The phenomenon was structural: once the Q-values associated with the boundary-crossing pattern exceeded those of genuine room completion, the replay buffer became saturated with these suboptimal transitions, and the agent could not recover without manual buffer flushing. Additionally, the exploration schedule required an unusually long fraction (60% of total steps) to be maintained in order to delay premature convergence to the exploitative policy.

B. PPO Configuration 1: Narrow Network

The narrow-network PPO configuration achieved initial learning within the first 3,000,000 steps without any reward shaping, demonstrating that PPO's on-policy rollout mechanism inherently provides richer gradient information than DQN's uniform replay. However, the narrow convolutional feature extractor (32→64→64) exhibited a pattern consistent with the findings of Hessel et al. [12]: insufficient representational capacity for the visually dense Venture environment led to policy instability after early room discoveries. The network struggled to disambiguate between visually similar states—particularly rooms with identical enemy positions but different treasure availability—causing the advantage estimates to oscillate. The result was a learning curve that showed initial progress followed by partial policy degradation: the agent learned to navigate hallways but failed to consolidate a robust room-clearing strategy. Return variance remained elevated, though substantially lower than the DQN baseline.

C. PPO Configuration 2: Wide Optimised Network

The `VentureOptimizedNetwork` (64→128→128) demonstrated markedly superior stability across all measured dimensions. The wider convolutional filters increased the spatial feature resolution available to the actor-critic heads, enabling finer discrimination between dungeon states. The training loss decreased more monotonically, and the policy loss—which in PPO should remain positive when the clipped constraint is active [4]—was consistently positive throughout training, confirming that the clipping mechanism was engaged and preventing destructive updates. Crucially, no reward exploitation events were observed: the critic's long-horizon value estimates correctly discounted the value of oscillatory behaviour that yields no treasure, guiding the policy toward genuine room completion. The resulting learning curve converged in fewer effective steps than both DQN and PPO-Narrow, with substantially reduced inter-episode variance.

The key mechanistic explanation for the stability improvement is the broader feature extractor's ability to encode longer spatial patterns. Venture dungeons require the agent to track its own position relative to room entrances, monster trajectories, and treasure locations simultaneously. A 64→128→128 network provides sufficient channel capacity for this multi-object tracking, reducing the variance in advantage estimates and thereby reducing the variance of the policy gradient [10], [12].

D. Absence of Reward Exploitation in PPO

The absence of reward exploitation in both PPO configurations is the most theoretically significant finding of this work. DQN's exploitation arose from a specific interaction between the off-policy replay buffer and the shaped reward manifold: because transitions from exploitative episodes were stored and re-sampled, the gradient associated with boundary-crossing outweighed the gradient from genuine treasure collection. PPO's on-policy architecture eliminates this failure mode structurally. Each rollout provides an unbiased sample from the current policy's state distribution; if the current policy is exploiting a boundary-crossing pattern, the value critic quickly learns that such trajectories do not yield the high discounted return R_t associated with full room completion. The advantage estimate for boundary-crossing transitions therefore becomes negative, discouraging the behaviour in the next policy update [4], [6]. This self-correcting property is absent from DQN because stored exploitative transitions continue to generate positive TD errors long after the exploitative behaviour has been eliminated from the current policy.

VII. Ablation Analyses

Network width (Narrow vs. Wide). The comparison between the 32→64→64 and 64→128→128 configurations directly isolates the effect of representational capacity. The wide network achieved lower return variance and earlier stable convergence, confirming that feature-extraction bandwidth is a binding constraint in Venture's visually complex state space. This result aligns with the Rainbow ablation study [12], which identifies insufficient network capacity as a primary cause of policy instability on hard Atari games.

Entropy coefficient. A preliminary sweep over $c_2 \in \{0.001, 0.01, 0.02\}$ showed that $c_2 = 0.001$ led to premature entropy collapse within the first 100,000 steps, yielding a degenerate policy that repeated a single action. $c_2 = 0.02$ maintained exploration effectively but reduced the signal-to-noise ratio of the policy gradient. $c_2 = 0.01$ provided the best trade-off, consistent with the recommendations for hard-exploration Atari games from Andrychowicz et al. [**?**] and the ICLR 2024 survey [8].

Horizon length. Reducing the rollout horizon to $T = 512$ resulted in the GAE estimate being dominated by bootstrap values, as most trajectories terminated before any reward event. $T = 1{,}024$ improved over $T = 512$ but still exhibited elevated loss variance due to insufficient positive-reward coverage per rollout. $T = 2{,}048$ provided the most stable gradients, corroborating the theoretical argument that the horizon must exceed the expected steps- to-first-reward [5].

Reward shaping in PPO. Unlike DQN, PPO did not require any form of RAM-assisted reward shaping to produce meaningful learning. This confirms that PPO's entropy-driven exploration is a functional substitute for domain-specific reward engineering in the context of ALE/Venture-v5, and supports the claim that the DQN shaping requirement was a consequence of gradient starvation rather than an intrinsic property of the environment.

VIII. Conclusions

This paper presented a systematic comparison of PPO and DQN on the ALE/Venture-v5 sparse-reward benchmark under a fixed computational budget. The key empirical finding is unambiguous: PPO converges faster, achieves higher and more stable episodic returns, and exhibits none of the reward-exploitation behaviour that characterised DQN training. These results are explained by three structural properties of PPO: (1) on-policy rollout collection eliminates the replay buffer lag that slows DQN adaptation; (2) the clipped surrogate objective prevents destructive large-update events triggered by sparse reward discoveries; and (3) the entropy bonus maintains stochastic exploration without the need for an explicit ε-greedy decay schedule.

The ablation study further demonstrated that network width is a critical design choice in visually complex environments: the 64→128→128 configuration significantly outperformed the standard 32→64→64 baseline in terms of stability and convergence, while also requiring no reward shaping.

Limitations and future work: the current study is limited to two PPO configurations and a single random seed due to computational constraints. Future work should replicate findings across at least three seeds and explore the integration of Double DQN [12] and Prioritized Experience Replay [14] into the DQN baseline to narrow the performance gap. Additionally, combining PPO's on-policy stability with Random Network Distillation (RND) [3] as an intrinsic curiosity signal represents a promising direction for further exploration of the Venture dungeon.

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